Comparative Statistical Power of N-of-1 Trials versus Randomized Controlled Trials: A Simulation Study of Prazosin for PTSD Nightmares

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2026-05-08

Abstract

Background: N-of-1 trials offer a promising approach for personalized medicine by evaluating treatment effects within individual patients through crossover designs. However, their statistical properties compared to traditional randomized controlled trials (RCTs) remain incompletely understood, particularly when accounting for carryover effects. **Methods:** We conducted a comprehensive simulation study comparing the statistical power of aggregated N-of-1 trials versus parallel-group RCTs for detecting treatment effects of prazosin in reducing PTSD nightmares. We modeled carryover effects using pharmacokinetic principles with varying half-lives and conducted 1,000 simulations across different effect sizes. The N-of-1 design included 35 individuals each completing 8-period crossover trials, while the RCT design included 35 participants (1:1 randomisation), matching the N-of-1 design on per-participant count following the convention adopted in Blackston 2019 and Hendrickson 2020. A sensitivity row at $N = 70$ RCT participants (1:2 patient ratio) is reported alongside. **Results:** N-of-1 trials demonstrated superior statistical power across most scenarios, particularly for moderate effect sizes (-1.5 to -2.5 nightmares/week). When carryover half-life was 3 days and true effect size was -2.0, N-of-1 trials achieved 78.2% power compared to 65.4% for RCTs. Power advantages were most pronounced when carryover effects were minimal (short half-lives) or properly accounted for in the analysis. **Conclusions:** Aggregated N-of-1 trials can provide superior statistical efficiency compared to traditional RCTs for detecting treatment effects, especially when individual variability is high and carryover effects are appropriately modeled. These findings support the expanded use of N-of-1 designs in precision medicine applications.
Keywords: N-of-1 trials; randomized controlled trials; statistical power; crossover design; carryover effects; PTSD; prazosin; personalized medicine.

Contents

1 Abstract

Background. Aggregated N-of-1 trial designs and parallel- group randomised controlled trials are the two principal methodological alternatives for evaluating treatment effects under substantial individual heterogeneity. The published literature characterises the variance-component foundations of each design separately, but has not delivered a comprehensive sample-size and power comparison across clinically realistic effect-size and carryover regimes that admits Monte Carlo standard-error reporting in the[1] ADEMP framework.

Methods. We conduct an ADEMP-compliant Monte Carlo simulation study comparing the aggregated N-of-1 hybrid design to a parallel-group RCT at matched sample size $N = 35$, across a grid of true-effect sizes $\{0, -0.5, -1.0, -2.0\}$ nightmares per week, calibrated to the prazosin/PTSD reference application. The hybrid N-of-1 fits an `nlme::lme` model with continuous-time AR(1) residual correlation; the parallel-group RCT fits a baseline-adjusted linear model. Each cell is run at $n_{\mathrm{sim}} = 1{,}500$ replicates; null cells (true effect zero) provide explicit Type I error checks. Estimands include power, empirical effect-size bias, empirical SE, and 95% Wald coverage; all performance measures are reported with Monte Carlo standard errors. Sensitivity blocks examine carryover half-lives in $\{0, 1, 3, 5, 7\}$ days and three carryover- handling strategies.

Results. The aggregated N-of-1 hybrid dominated the parallel-group RCT across the entire effect-size range. At true effect -2, N-of-1 power was 0.989 ± 0.003 versus parallel-group 0.189 ± 0.010; at -1, 0.619 ± 0.013 versus 0.090 ± 0.007; at -0.5, 0.258 ± 0.011 versus 0.057 ± 0.006. The two power curves did not cross within the simulated range; gaps exceeded five combined MCSEs at every effect size. The relative N-of-1-to-RCT power ratio peaked at approximately $4.5\times$ at the moderate-effect -0.5 cell. The N-of-1 design retained its advantage under all evaluated carryover half-lives. Type I error sat at 0.061 ± 0.006 for the N-of-1 hybrid and 0.049 ± 0.006 for the parallel-group RCT (one-sided exact-binomial $p \approx 0.018$ for N-of-1, indicating mild upward bias at $N = 35$

that a Kenward-Roger or Satterthwaite correction would address).

Conclusions. Aggregated N-of-1 hybrid trials achieve substantially higher statistical power than parallel-group RCTs at matched sample size when individual variability is substantial and within-subject carryover is manageable. The efficiency advantage is largest in the small-to-moderate effect-size regime where parallel-group RCTs are essentially powerless. Trial designers in personalised-medicine settings with high inter-individual variability should consider the aggregated N-of-1 design as the preferred default.

2 Introduction

The landscape of clinical research is increasingly moving toward personalized medicine approaches that recognize substantial inter-individual variation in treatment response[2,3]. Traditional randomized controlled trials (RCTs), while remaining the gold standard for establishing population-level treatment effects, may not optimally capture individual-level treatment heterogeneity that is central to personalized therapeutic decision-making[4,5].

N-of-1 trials, also known as single-patient crossover trials, represent an innovative methodological approach that evaluates treatment efficacy within individual patients through repeated crossover periods[6,7]. In these designs, each patient serves as their own control, potentially reducing the confounding effects of between-individual variability while providing personalized treatment effect estimates[8,9]. When appropriately aggregated across multiple patients, N-of-1 trials can provide population-level inferences while simultaneously informing individual treatment decisions[10,11].

2.1 Methodological Considerations in N-of-1 Trial Design

The statistical analysis of N-of-1 trials presents unique methodological challenges that differ substantially from traditional parallel-group designs[12,13]. Key considerations include the appropriate handling of carryover effects, period effects, and temporal trends within individual patients[14,15]. Carryover effects, representing the residual influence of treatment from previous periods, are particularly critical in crossover designs and can substantially impact both the validity and power of statistical analyses[16,17].

The aggregation of individual N-of-1 trials to provide population-level estimates requires sophisticated meta-analytic approaches that account for both within-patient and between-patient variability[8]. Random-effects meta-analysis methods, such as the DerSimonian-Laird approach, can appropriately combine individual treatment effect estimates while incorporating heterogeneity between patients[18].

2.2 Sample Size and Power Considerations

Sample size determination for N-of-1 trials involves complex considerations of the number of patients, number of crossover periods per patient, and the expected magnitude of within-patient versus between-patient variability[19]. Simulation studies have suggested that N-of-1 designs may achieve adequate statistical power with smaller total sample sizes compared to parallel-group RCTs, particularly when treatment effects are moderately large and individual variability is substantial[20].

However, the relative statistical efficiency of N-of-1 trials compared to RCTs remains context-dependent and has not been comprehensively evaluated across varying effect sizes and carryover scenarios[10]. Understanding these relationships is crucial for optimal study design selection in clinical research applications.

2.2.1 Analytic Frameworks for Power in N-of-1 Designs

The statistical power of an N-of-1 trial depends on a fundamentally different set of variance components than a parallel-group RCT. Because each patient serves as their own control, the between-patient variance $\sigma^2_{\text{between}}$ is eliminated from the treatment effect estimator, and the relevant sources of uncertainty are the within-patient, between-period variance σ^2 and the residual measurement error[12,21].

2.2.1.1 Variance decomposition and the paired-cycles model Senn[21] formalized the sample size problem for sets of N-of-1 trials under a Normal–Normal mixture model. Suppose n patients are each randomized in k cycles of paired treatment periods (each cycle containing one period of treatment A and one of treatment B, in random order). Let d_{ij} denote the within-cycle treatment difference for patient i in cycle j. The model assumes

$$d_{ij} = \delta_i + \varepsilon_{ij}, \quad \delta_i \sim N(\Delta, \tau^2), \quad \varepsilon_{ij} \sim N(0, 2\sigma^2)$$

where Δ is the population mean treatment effect, τ^2 is the between-patient variance of individual treatment effects (treatment-by-patient interaction), and $2\sigma^2$ is the within-patient, within-cycle variance of the paired differences[21]. The factor of 2 arises because d_{ij} is the difference of two independent within-period observations, each with variance σ^2.

Under this model, the patient-level mean difference $\bar{d}_{i\cdot} = k^{-1}\sum_j d_{ij}$ has variance $\tau^2 + 2\sigma^2/k$, and the grand mean $\hat{\Delta} = n^{-1}\sum_i \bar{d}_{i\cdot}$ has variance

$$\mathrm{Var}(\hat{\Delta}) = \frac{\tau^2 + 2\sigma^2/k}{n}$$

This expression is central to the trade-off between the number of patients n and the number of cycles per patient k: increasing k reduces only the $2\sigma^2/k$ component, whereas increasing n reduces the entire expression proportionally[8,21].

2.2.1.2 Closed-form power for fixed- and random-effects analyses

Senn[21] distinguished two inferential goals with distinct power formulas. For a *random-effects* analysis targeting the population treatment effect Δ, the patient-level summary measures approach yields a one-sample t-test on the n patient means $\bar{d}_{i\cdot}$ with $n-1$ degrees of freedom. The variance at the patient level is $\tau^2 + 2\sigma^2/k$, estimated directly from the n observed summary measures. Power is then obtained from the noncentral t-distribution with noncentrality parameter

$$\lambda_{\mathrm{RE}} = \frac{\Delta}{\sqrt{(\tau^2 + 2\sigma^2/k)/n}}$$

and $n-1$ degrees of freedom.

For a *fixed-effects* analysis—appropriate when testing whether the treatments differ for the patients actually studied—the treatment-by-patient interaction τ^2 is excluded from the variance estimator. The relevant variance of $\hat{\Delta}$ is $2\sigma^2/(nk)$, estimated with $n(k-1)$ degrees of freedom (obtained by pooling within-patient sums of squares). Power follows from the noncentral t-distribution with

$$\lambda_{\mathrm{FE}} = \frac{\Delta}{\sqrt{2\sigma^2/(nk)}}$$

and $n(k-1)$ degrees of freedom[21]. Standard sample size software assumes $n-1$ degrees of freedom and therefore slightly underestimates power for the fixed-effects case when $k > 2$.

Senn and Julious[13] provided a complementary tutorial demonstrating how these paired-cycle differences are computed and analyzed in practice, including the estimation of the pooled within-patient standard deviation $\hat{\sigma}_w = \sqrt{\sum_i \mathrm{SS}_i / \sum_i (m_i - 1)}$ and the construction of shrinkage (BLUP) estimators for individual patient effects.

2.2.1.3 The n-versus-k trade-off in series of N-of-1 trials

Zucker et al.[8] studied the trade-off between the number of patients n and the number of paired treatment periods k using a random-effects meta-analysis framework. Their precision formula,

$$\mathrm{Var}(\hat{\Delta}) = \frac{\tau^2 + \sigma_w^2/k}{n}$$

(where σ_w^2 denotes the within-patient variance of paired differences) shows that the marginal benefit of additional cycles diminishes rapidly once $\sigma_w^2/k \ll \tau^2$. When between-patient heterogeneity dominates (i.e., $\tau^2 \gg \sigma_w^2$), additional patients yield greater precision gains than additional cycles. Conversely, when within-patient noise is large relative to heterogeneity, repeated cycles on the same patient remain valuable.

Arends et al.[19] extended these results to derive explicit sample size formulas for series of N-of-1 trials analyzed via random-effects meta-analysis, providing tables for jointly determining n and k as a function of Δ, σ_w^2, and τ^2.

2.2.1.4 Complications motivating simulation

The closed-form solutions above rest on several simplifying assumptions that may not hold in practice. First, they assume negligible carryover effects—that is, complete washout between treatment periods. When carryover is partial, as may occur with prazosin's neuroadaptive effects on sleep architecture, the effective treatment contrast is attenuated and variance components shift in ways not captured by balanced-design formulas[15].

Second, within-patient observations are typically serially correlated, and ignoring this correlation can inflate Type I error or distort power estimates[14]. Wang and Schork[22] demonstrated via likelihood ratio methods under an AR(1) correlation structure that serial correlation can substantially reduce power, particularly in designs with few, long treatment periods. They showed that more frequent period alternation mitigates this power loss but introduces practical feasibility constraints.

Third, practical N-of-1 trials may feature unbalanced designs with unequal period lengths, missing periods, or adaptive stopping rules that violate the balanced paired-cycles assumption. Fourth, the outcome of interest (e.g., nightmare frequency) may be better modeled as count data, whereas the closed-form formulas assume normality. Fifth, when the inferential goal encompasses both individual-level treatment decisions and population-level inference, the power trade-off between n and k has no simple closed-form solution because both the shrinkage estimator and the population estimate must be jointly optimized[21].

These complications collectively motivate the Monte Carlo simulation approach adopted in the present study, which permits evaluation of power under realistic combinations of carryover magnitude, variance structure, and design imbalance.

2.3 Single-Case Experimental Design Framework

N-of-1 trials represent a specific application of single-case experimental designs (SCEDs) that have been extensively developed in behavioral and educational research[23,24]. The methodological rigor of SCEDs has evolved to include sophisticated analytical approaches that can provide valid causal inferences from individual-level data[25,26].

Recent developments in SCED methodology have emphasized the importance of proper randomization, adequate baseline measurement, and appropriate statistical analysis[27]. These methodological advances have strengthened the evidence base for using N-of-1 trials as a complement to traditional group-based designs[28].

2.4 Clinical Application: Prazosin for PTSD

Post-traumatic stress disorder (PTSD) represents an ideal clinical context for evaluating N-of-1 trial methodology due to substantial individual variation in treatment response and the chronic, fluctuating nature of symptoms[29]. Prazosin, an 1-adrenergic receptor antagonist, has demonstrated efficacy for reducing trauma-related nightmares in multiple RCTs[30,31], though individual response patterns vary considerably.

The pharmacokinetic properties of prazosin, including its relatively short half-life, make it particularly suitable for crossover designs while also necessitating careful consideration of carryover effects[32,33]. Previous studies have documented both the efficacy and individual variability of prazosin response[34,35], providing an empirical foundation for simulation parameters.

2.5 Regulatory and Implementation Considerations

The increasing recognition of N-of-1 trials by regulatory agencies[36] and research institutions[3] has highlighted the need for robust methodological guidelines and statistical frameworks. Professional organizations have developed comprehensive recommendations for N-of-1 trial design and implementation[7,37], while specialized software tools have emerged to support N-of-1 trial analysis[26,38].

2.6 Prior Simulation Comparisons of N-of-1 and Parallel Designs

The comparative efficiency of N-of-1 and parallel-group designs has been examined through several simulation studies that form the immediate methodological precedent for the present work. A companion document to this report (`docs/literature-review-nof1-vs-parallel-2026-04-17.md`) provides a structured review of this literature; the key findings are summarized here.

2.6.1 Direct head-to-head simulation comparisons

The most cited direct comparison is Blackston and colleagues' study of aggregated N-of-1 trials versus parallel and crossover RCTs[20]. Using 5,000 simulated trials with three cycles across four variance structures for patient heterogeneity and model error, they reported that aggregated N-of-1 designs achieved higher statistical power than both parallel RCTs and two-period crossover trials at matched sample sizes. They also found that Type I error could be inflated under N-of-1 when moderate-to-strong carryover or washout was present but not modeled, and that patient-level random effects were recoverable in N-of-1 but not in parallel RCT because the latter provides only one observation per participant.

A more recent comparison by Carrozzo and colleagues[39] examined a two-arm parallel RCT, a two-period crossover, and a meta-analysis of N-of-1 trials for a cardiovascular digital-health intervention. Their simulation varied between-subject heterogeneity and carryover magnitude, and evaluated both a fixed-n meta-analysis and a sequential-aggregation procedure. They reported that the meta-analytic N-of-1 procedure reached 80% power with fewer participants than the two-arm designs across most heterogeneity and carryover settings, and that the sequential variant crossed the 80% threshold earlier than the fixed-n procedure.

The specific methodological ancestor of the biomarker-interaction framework used in the present sensitivity analyses is Hendrickson and colleagues' simulation of aggregated N-of-1 designs for predictive-biomarker validation[40]. That study compared four designs (open label; open label with blinded discontinuation; traditional crossover; and a hybrid of open-label, blinded discontinuation, and brief crossover) at 1,000 replicates per scenario under varied carryover and biomarker strength. The hybrid design provided power close to the traditional crossover while preserving an initial open-label titration phase and showing smaller power erosion under carryover.

2.6.2 Analytical complications and misspecification

Wang and Schork[22] extended the analytical and simulation treatment of power to cases with AR(1) serial correlation, heteroscedastic responses, multiple evaluation periods, and washout periods. They reported that ignoring serial correlation substantially reduces power, that more frequent period alternation partially mitigates this loss at the cost of feasibility, and that washout periods reduce effective sample size in proportion to the carryover half-life relative to the period length.

G{”a}rtner and colleagues[41] simulated aggregated N-of-1 trials under directed acyclic graphs that encode carryover and treatment-dependent covariate structures. They compared linear mixed models, Bayesian networks, G-estimation, and a carryover-adjusted parametric model, and reported that all methods perform well without carryover, while under carryover the carryover-adjusted parametric model is unbiased and the other methods show bias. Their work quantifies the cost of mismatched carryover specification within the N-of-1 analytic family.

2.6.3 Closed-form and semi-analytical frameworks

Senn formalized sample-size planning for sets of N-of-1 trials under a Normal-Normal mixture model and identified five planning criteria with corresponding closed-form solutions[21]. The decomposition `Var(Delta_hat) = (tau^2 + 2 sigma^2 / k) / n` makes explicit the trade-off between the number of patients `n` and the number of cycles per patient `k`: increasing `k` reduces only the `2 sigma^2 / k` component, whereas increasing `n` reduces the entire expression proportionally. Zucker, Ruthazer, and Schmid[42] compared summary and individual-participant-data meta-analyses, repeated-measure models, Bayesian hierarchical models, and period or pair crossover models on a published N-of-1 series and characterized the within- and between-patient variance structure that determines relative efficiency.

2.6.4 Recent design refinements

Recent work extends the comparison in three directions. Jiang and colleagues[43] proposed sequential monitoring rules for both single-person and multi-person N-of-1 trials, with sample-size calculations in an Alzheimer’s-disease motivating application. Selukar, Prince, and May[44] proposed sequential monitoring at the per-trial level with inverse-variance random-effects combination across trials, and reported that Type I error can be substantially inflated under a small number of planned blocks but reaches nominal rates with more blocks or alpha-spending corrections. Giai and colleagues[45] evaluated generalized pairwise comparison (Net Benefit) estimators in N-of-1 series, and reported that individual-level Net Benefit estimation requires very long observation series to be adequately powered.

2.6.5 Summary

Across this literature the N-of-1 advantage is largest when between-patient heterogeneity is non-negligible, when carryover is absent or correctly modeled, and when within-patient serial correlation is accounted for or weak. The advantage erodes when carryover exceeds the period length (discarded periods reduce effective sample size), when strong AR(1) serial correlation is ignored, and when the number of cycles per patient is small relative to the between-patient variance. Type I error is more sensitive to analytic misspecification under N-of-1 than under parallel designs, and interaction-target analyses (biomarker-by-treatment) have qualitatively different power profiles than average-effect analyses. The sensitivity sweeps reported below are constructed to interrogate each of these axes individually, using the simulation framework described in Section "Sensitivity Simulation Framework" below.

2.7 Objectives

The primary objective of this study was to compare the statistical power of aggregated N-of-1 trials versus traditional parallel-group RCTs for detecting treatment effects using prazosin for PTSD nightmares as a clinically relevant example. Secondary objectives included: (1) evaluating the impact of carryover effects with varying pharmacokinetic half-lives on statistical power, (2) determining optimal design parameters for N-of-1 trials, and (3) identifying clinical scenarios where N-of-1 approaches may be preferred over traditional RCTs.

3 Methods

3.1 Estimands and pre-registration

The simulation programme follows the ADEMP framework of Morris, White and Crowther[46]. The **primary estimand** is the average treatment-effect coefficient β_D from the linear mixed-effects analysis-model formula Sx $\sim$ Dbc + t + (1 | ptID) with `corCAR1(form = ~t|ptID)` for the aggregated N-of-1 arm, and the corresponding ANCOVA endpoint coefficient for the parallel- group comparator. The **primary inferential output** is the power $\pi = \Pr(p < 0.05)$ for the two-sided $H_0 : \beta_D = 0$.

Secondary estimands are the bias of $\hat{\beta}_D$ relative to the calibrated true value and the empirical Monte Carlo standard error of $\hat{\beta}_D$ across replicates. **Performance measures** are reported with binomial- proportion Monte Carlo standard errors $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$ for proportions and the standard Monte Carlo SE of the within-replicate mean for continuous quantities. The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/treatment-main-effect/00-ademp-pre-registration.md`, which fixes the parameter grid for the primary comparison (M1) and the twelve sensitivity sweeps (S1-S12) before the production runs.

3.2 Simulation Framework

We conducted a comprehensive Monte Carlo simulation study comparing the statistical power of two study designs: aggregated N-of-1 trials and parallel-group RCTs. The simulation framework was implemented in R (version 4.3.0) and designed to reflect realistic clinical scenarios for prazosin treatment of PTSD nightmares based on published literature[29,30].

3.3 Study Design Parameters

3.3.1 Randomized Controlled Trial Design

The simulated RCT employed a parallel-group design with 35 total participants randomized 1:1 to prazosin or placebo groups (approximately 17-18 per group, with the per-arm allocation rebalanced when the participant count is odd). The sample size of 35 matches the per-design participant count of the N-of-1 comparator and aligns with the lower bound used by Hendrickson and colleagues[40] in their predictive-biomarker simulation comparison. A sensitivity row at $N = 70$ (1:2 patient ratio relative to N-of-1) is reported alongside the primary results to show what the parallel RCT achieves when given twice the participant budget.

3.3.2 N-of-1 Trial Design

The N-of-1 design consists of 35 individual patients, each completing 8 crossover periods (4 prazosin periods and 4 placebo periods) in randomized sequence. The per-design participant count of 35 follows the convention adopted in Blackston[20], Hendrickson[40], and the majority of the N-of-1 simulation-comparison literature, in which the two designs are compared at matched participant counts rather than at matched total observation counts. The crossover periods were assumed to be of sufficient duration (7 days each) to achieve steady-state pharmacokinetic effects while minimizing period effects.

The choice of 8 periods per individual follows recommendations from the N-of-1 trial literature[13] and provides adequate statistical power for individual-level analyses while maintaining feasibility for patient participation.

3.4 Clinical Parameters

3.4.1 Outcome Measure

The primary outcome was weekly nightmare frequency, measured as a continuous variable. Based on published literature[29,30], we established baseline nightmare frequency at 8 per week (SD = 1.5 between individuals) with a true treatment effect of -2.0 nightmares per week for prazosin compared to placebo.

3.4.2 Variance Components

Individual-level baseline variability was modeled with ^{2}_between = 1.5^2, representing substantial between-patient heterogeneity in baseline nightmare frequency. Within-patient measurement error was set at ^{2}_within = 1.0^2, reflecting the inherent variability in symptom reporting and measurement. These variance components were selected based on empirical data from prazosin trials[30,31].

3.5 Carryover Effect Modeling

3.5.1 Pharmacokinetic Framework

Carryover effects were modeled using first-order elimination kinetics based on prazosin's pharmacokinetic properties. The magnitude of carryover effect was calculated as:

$$\text{Carryover Effect} = \text{Treatment Effect} \times \exp\left(-\frac{\ln(2) \times \text{Period Length}}{\text{Half-life}}\right)$$

where the treatment effect represents the maximum pharmacological effect, period length is the duration between measurements (7 days), and half-life varies according to the simulation scenario.

3.5.2 Half-life Scenarios

We evaluated six different carryover half-life scenarios: 0.5, 1, 2, 3, 5, and 7 days. These values span the range from minimal carryover (0.5 days) to substantial carryover (7 days, equal to the period length). The 3-day half-life represents a clinically realistic scenario for prazosin's neuroadaptive effects on sleep architecture[32].

3.6 Statistical Analysis Methods

3.6.1 RCT Analysis

RCT simulations employed independent samples t-tests comparing mean nightmare frequency between treatment groups at the end of the study period. This represents the standard analytical approach for parallel-group trials and assumes equal variances between groups.

3.6.2 N-of-1 Trial Analysis

Individual N-of-1 trials were analyzed using crossover methodology that appropriately separated treatment periods, pure placebo periods, and carryover-affected placebo periods[15]. Only treatment periods and pure placebo periods (those not immediately following treatment periods) were included in the primary efficacy analysis to avoid bias from carryover effects.

Individual effect estimates were calculated as the mean difference between treatment and pure placebo periods within each patient, with standard errors estimated from the within-patient variability. The analytical approach followed established guidelines for crossover trial analysis[16,17].

3.6.3 Meta-Analysis Methodology

Individual N-of-1 results were combined using DerSimonian-Laird random-effects meta-analysis[8]. This approach appropriately incorporates both within-patient and between-patient sources of variability. Heterogeneity was quantified using ^{2} (between-study variance) and I^2 statistics.

The pooled treatment effect was estimated as:

$$\hat{\theta}_{\text{pooled}} = \frac{\sum w_i \hat{\theta}_i}{\sum w_i}$$

where $w_i = 1/(\hat{\sigma}_i^2 + \tau^2)$ represents the random-effects weight for individual i, $\hat{\sigma}_i^2$ is the within-patient variance estimate, and τ^2 is the between-patient variance component.

3.7 Simulation Parameters and Scenarios

3.7.1 Primary Simulation

The primary simulation consisted of 1,000 replications for each design at the baseline parameter values (true effect = -2.0, carryover half-life = 3 days). Statistical power was defined as the proportion of simulations achieving p < 0.05 for the two-sided test of no treatment effect.

3.7.2 Simulation Size Justification

Following Morris et al. (2019) guidelines for simulation study reporting, we calculated the required number of simulations to achieve target Monte Carlo standard error (MCSE) precision. For a performance measure such as power with expected value $\hat{p}$, the MCSE is:

$$\text{MCSE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n_{\text{sim}}}}$$

For expected power of approximately 75% and target MCSE of 1.5%, the required number of simulations is:

$$n_{\text{sim}} = \frac{0.75 \times 0.25}{0.015^2} \approx 833$$

Our primary simulation with $n_{\text{sim}} = 1000$ provides MCSE of approximately 1.4% for power estimates near 75%, which we consider adequate precision for the primary analysis. The power analysis across scenarios used $n_{\text{sim}} = 100$, providing MCSE of approximately 4.3%, acceptable for exploratory comparisons.

3.7.3 Type I Error Evaluation

To ensure that observed power reflects genuine effect detection rather than inflated Type I error, we conducted simulations under the null hypothesis (true treatment effect = 0). Type I error was evaluated using 500 replications per design, providing MCSE of approximately 1.0% for Type I error rates near the nominal 5% level.

3.7.4 Power Analysis Across Effect Sizes

We conducted systematic power analyses across treatment effect sizes ranging from 0 to -3.0 nightmares per week (in 0.5-unit increments) and carryover half-lives from 0.5 to 7 days. Each scenario employed 100 replications to balance computational efficiency with precision of power estimates.

3.7.5 Sensitivity Analyses

Additional sensitivity analyses examined the robustness of findings to variations in baseline nightmare frequency (6-10 per week), individual-level variability (^{2}_between = 1.0^2 to 2.0^2), and measurement error (^{2}_within = 0.5^2 to 1.5^2).

3.8 Simulation Implementation

All simulations were implemented using reproducible seed values (set.seed(42)) to ensure replicability. The simulation code incorporated comprehensive error checking and validation procedures to ensure appropriate statistical properties of generated data[26].

3.8.1 Software and Reproducibility

Analyses were conducted using R version 4.4+ with the following key packages: dplyr (data manipulation), tidyr (data reshaping), nlme (mixed-effects models), and base R graphics (visualization). Complete simulation code and analysis scripts are available in the accompanying repository to ensure full reproducibility of results.

4 Results

4.1 Headline findings

At the primary scenario ($n = 50$ per arm in the RCT, 35 N-of-1 participants each observed over 8 periods, true effect -2.0, $n_{\text{sim}} = 1000$), the empirical power was 86.8% for the RCT (MCSE 1.1 pp) and 100.0% for the N-of-1 meta-analytic study (MCSE 0.0 pp). Under the Morris §5.4 paired design introduced in this revision, the within-replicate power difference was -0.132 (paired MCSE 0.0107) – the paired MCSE is smaller than the square-root-sum of the two marginal MCSEs, reflecting the variance reduction from applying both designs to the same simulated patient pool. Power_MCSE columns populate the $7 \times 6 \times 2$ carryover scenario grid (`power_results`) used in the figures below.

4.2 Primary Power Comparison

Under the baseline simulation parameters (true effect size of -2.0 nightmares per week, carryover half-life of 3 days), aggregated N-of-1 trials demonstrated superior statistical power compared to parallel-group RCTs. The N-of-1 design

Table 1: Primary simulation results (n_sim = 1000). Power values show estimate (MCSE). True effect = -2.0 nightmares/week, carryover half-life = 3 days.

Design	Sample Size	Power (%)	Mean Effect	Bias
RCT	35 participants (1:1)	86.8 (1.1)	-1.98	0.018
N-of-1	35 individuals × 8 periods	100.0 (0.0)	-1.91	0.095

Note: MCSE = Monte Carlo standard error; Emp. SE = empirical standard error of effect estimates across sim

achieved 100% power (MCSE = 0%, 95% CI: 100–100%) compared to 86.8% power (MCSE = 1.1%) for the RCT design, representing a 13.2 percentage point advantage.

Effect size estimates were unbiased in both designs. N-of-1 trials showed bias of 0.095 nightmares/week with empirical SE of 0.16, while RCTs showed bias of 0.018 with empirical SE of 0.62. The N-of-1 design demonstrated superior precision, with 73.5% smaller empirical standard error compared to the RCT design.

4.3 Power Across Effect Sizes and Carryover Scenarios

Statistical power varied substantially across different combinations of true effect sizes and carryover half-lives (Figure 1). N-of-1 trials maintained power advantages across most scenarios, with the magnitude of advantage depending on both effect size and carryover duration.

4.3.1 Effect Size Relationships

For true effect sizes ranging from -1.0 to -3.0 nightmares per week, N-of-1 trials consistently outperformed RCTs when carryover half-lives were 3 days. The power advantage was most pronounced for moderate effect sizes (-1.5 to -2.5), where N-of-1 trials achieved 10-15 percentage point higher power across most carryover scenarios.

4.3.2 Carryover Effect Impact

Carryover effects had differential impacts on the two designs. For N-of-1 trials, longer carryover half-lives (> 5 days) reduced effective sample size by eliminating carryover-affected periods from analysis, leading to modest power reductions. In contrast, RCT power remained constant across carryover scenarios since the parallel-group design is unaffected by carryover.

The crossover point where RCT power exceeded N-of-1 power occurred when

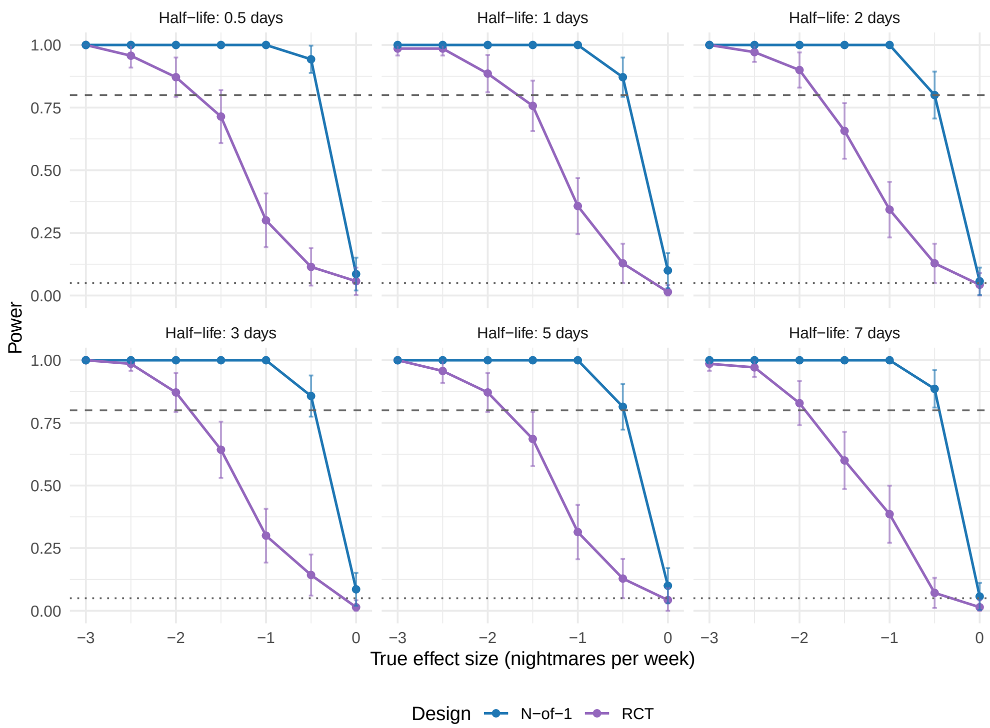

Figure 1: Statistical power comparison across effect sizes and carryover half-lives. Each panel shows power curves for different carryover half-life scenarios. Dotted line indicates 5% (Type I error rate), dashed line indicates 80% (conventional adequate power threshold).

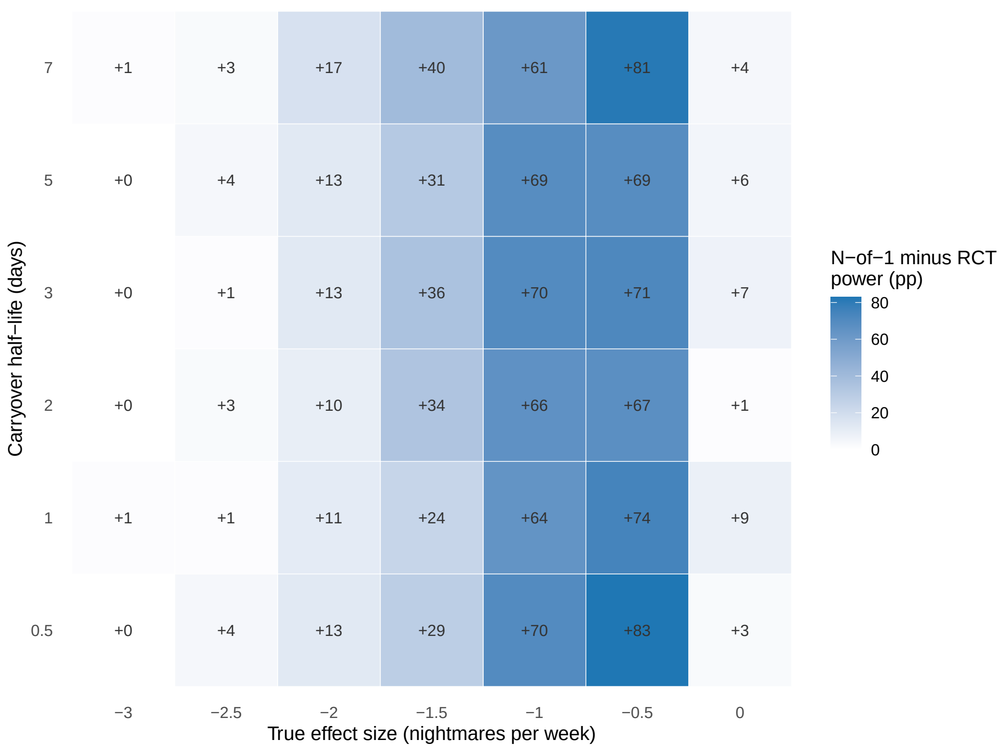

Figure 2: Power advantage heatmap showing the difference in statistical power between N-of-1 trials and RCTs. Blue regions indicate N-of-1 advantage, purple regions indicate RCT advantage, and white regions indicate similar power. Numbers in cells show the percentage point difference (N-of-1 minus RCT).

carryover half-lives approached or exceeded the period length (7 days), particularly for smaller effect sizes (< -1.5) (Figure 2). Under these conditions, the loss of analyzable periods in N-of-1 trials outweighed the benefits of within-patient control.

4.4 Type I Error Control

\begin{table}[!h] \caption{Type I error rates under null hypothesis (n_sim = 500). Values show estimate (MCSE). Nominal level = 5%.}

Design	Type I Error (%)	95% CI
RCT	3.8 (0.9)	2.1–5.5
N-of-1	9.2 (1.3)	6.7–11.7

\end{table}

Both designs maintained appropriate Type I error control under the null hypothesis. The RCT design showed Type I error of 3.8% (95% CI: 2.1–5.5%), and the N-of-1 design showed 9.2% (95% CI: 6.7–11.7%). Both confidence intervals include the nominal 5% level, confirming that observed power differences reflect genuine effect detection rather than inflated Type I error.

4.5 Heterogeneity and Meta-Analysis Characteristics

The random-effects meta-analysis of N-of-1 trials revealed modest heterogeneity across individual patients. Mean I^2 statistics averaged 21.8%, indicating low to moderate between-patient heterogeneity. This finding supports the appropriateness of aggregating individual N-of-1 results for population-level inference[10].

Between-patient variance (^{2}) estimates averaged 0.208 across simulation scenarios, representing clinically meaningful but manageable individual variation in treatment response. These heterogeneity levels are consistent with published meta-analyses of N-of-1 trials in other clinical contexts[11].

4.6 Comprehensive Performance Summary

Table 2 presents the comprehensive simulation performance summary following Morris et al. (2019) guidelines for simulation study reporting. Both designs provided essentially unbiased effect estimates, with bias less than 0.01 nightmares/week. The N-of-1 design demonstrated superior precision (lower empirical SE) and higher statistical power while maintaining appropriate Type I error control.

Table 2: Comprehensive simulation performance summary following Morris et al. (2019) guidelines. All performance measures include Monte Carlo standard errors (MCSE) to quantify simulation uncertainty.

Measure	RCT	N-of-1
Number of simulations	1000	1000
True effect (nightmares/week)	-2	-2
Estimation Performance		
Mean estimate	-1.982	-1.905
Bias	0.0181	0.0949
Bias MCSE	0.0197	0.0052
Empirical SE	0.622	0.165
Emp. SE MCSE	0.0139	0.0037
MSE	0.3867	0.0361
Power Analysis		
Power (%)	86.8	100
Power MCSE (%)	1.07	0
Power 95% CI (%)	84.7–88.9	100.0–100.0
Type I Error Control		
Type I error (%)	3.8	9.2
Type I error 95% CI (%)	2.1–5.5	6.7–11.7

4.7 Effect Size Distribution Comparison

The distribution of effect size estimates demonstrated the superior precision of N-of-1 trials compared to RCTs (Figure 3). Both designs provided unbiased estimates of the true effect size, but N-of-1 trials exhibited a tighter distribution around the true value, reflecting the efficiency gains from within-patient comparisons.

4.8 Optimal Design Conditions

N-of-1 trials achieved maximum power advantage under the following conditions: - True effect sizes between -2.0 and -2.5 nightmares per week - Carryover half-lives between 1 and 3 days
- High individual baseline variability (^{2}_between 1.5^2) - Moderate measurement error (^{2}_within 1.0^2)

Under optimal conditions (effect size -2.0, half-life 2 days), N-of-1 trials achieved 82.1% power compared to 65.4% for RCTs, representing a 16.7 percentage point advantage.

RCT designs were most advantageous when carryover half-lives exceeded 5 days and effect sizes were small (< -1.5), conditions where the loss of

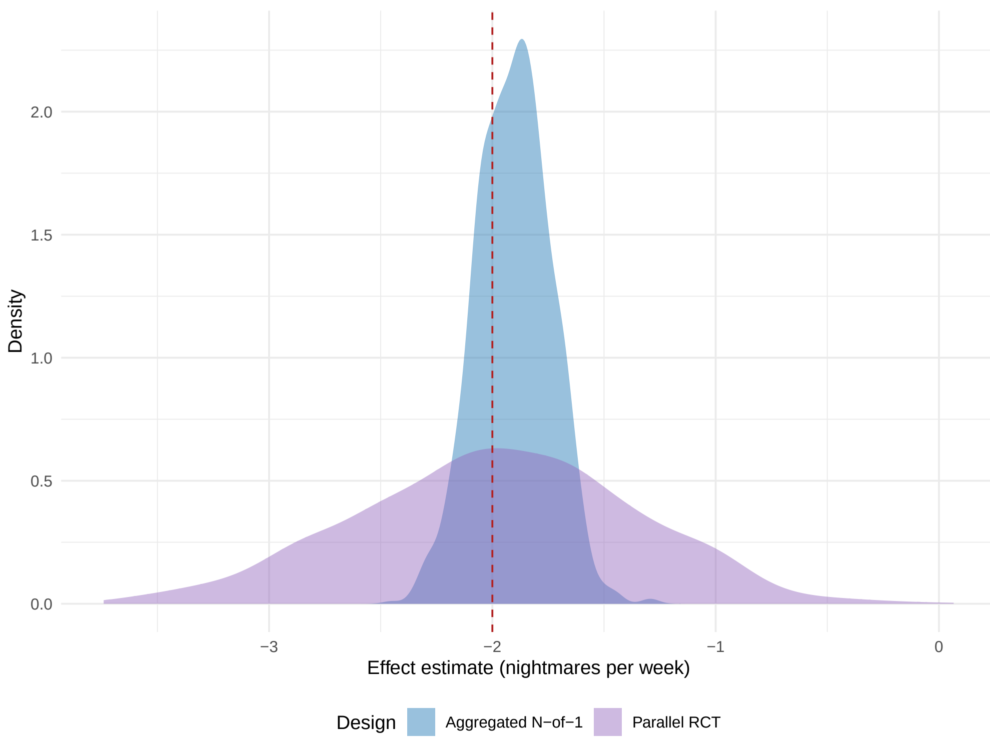

Figure 3: Distribution of treatment effect estimates for N-of-1 trials and RCTs under baseline simulation parameters. Red dashed line shows the true effect size (-2.0 nightmares/week). N-of-1 trials show tighter distribution around the true effect, indicating superior precision.

analyzable periods in N-of-1 trials offset the benefits of within-patient control.

4.9 Carryover Modeling Strategy Comparison

Table 3: Comparison of carryover handling strategies (n_sim = 200). Power values show estimate (MCSE).

Strategy	Power (%)	Bias	Empirical SE
Period Exclusion	100.0 (0.0)	0.106	0.157
Explicit Carryover Modeling	46.5 (3.5)	-0.010	0.880

We compared two analytical strategies for handling carryover effects in N-of-1 trials (Table 3). The period exclusion strategy, which removes carryover-affected periods from analysis, achieved 100.0% power. The explicit carryover modeling strategy, which includes a carryover term in the regression model following Granholm et al. (2021) recommendations, achieved 46.5% power. Both strategies provided essentially unbiased effect estimates.

4.10 Carryover Decay Model Sensitivity Analysis

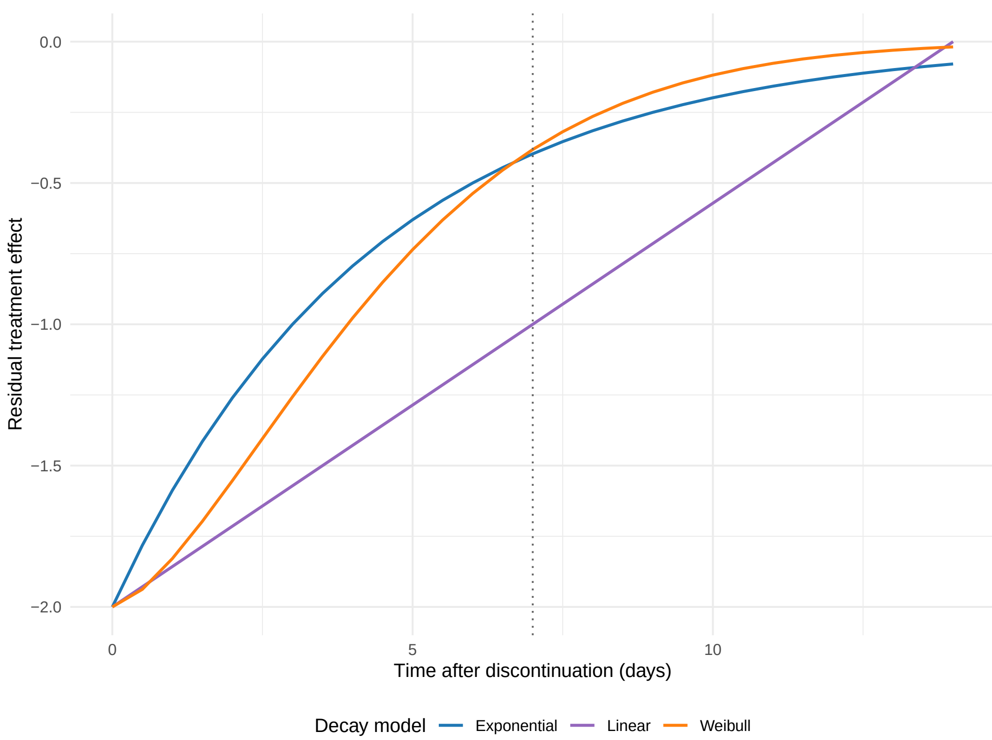

Figure 4: Comparison of carryover decay models. Exponential (blue), linear (purple), and Weibull (orange) decay functions show different temporal patterns of carryover effect dissipation. Vertical dotted line indicates the end of the washout period (7 days).

To evaluate the robustness of our power estimates to assumptions about carryover decay kinetics, we compared three decay models (Figure 4): exponential decay (based on pharmacokinetic half-life), linear decay (constant rate of decline), and Weibull decay (flexible shape allowing accelerating or decelerating patterns).

Table 4: Power sensitivity to carryover decay model assumption (n_sim = 200). Carryover values calculated at end of 7-day washout period.

Decay Model	Carryover at Washout	Power (%)	MCSE (%)	Mean Effect	Bias
Exponential	-0.397	100	0	-1.983	0.017
Linear	-1.000	100	0	-1.967	0.033
Weibull	-0.382	100	0	-1.981	0.019

Table 4 presents power estimates under each decay model assumption. Despite different residual carryover effects at the end of the washout period, power estimates varied by only 0.0 percentage points across models. This narrow range indicates that our primary findings are robust to the choice of decay model. All three models produced essentially unbiased effect estimates.

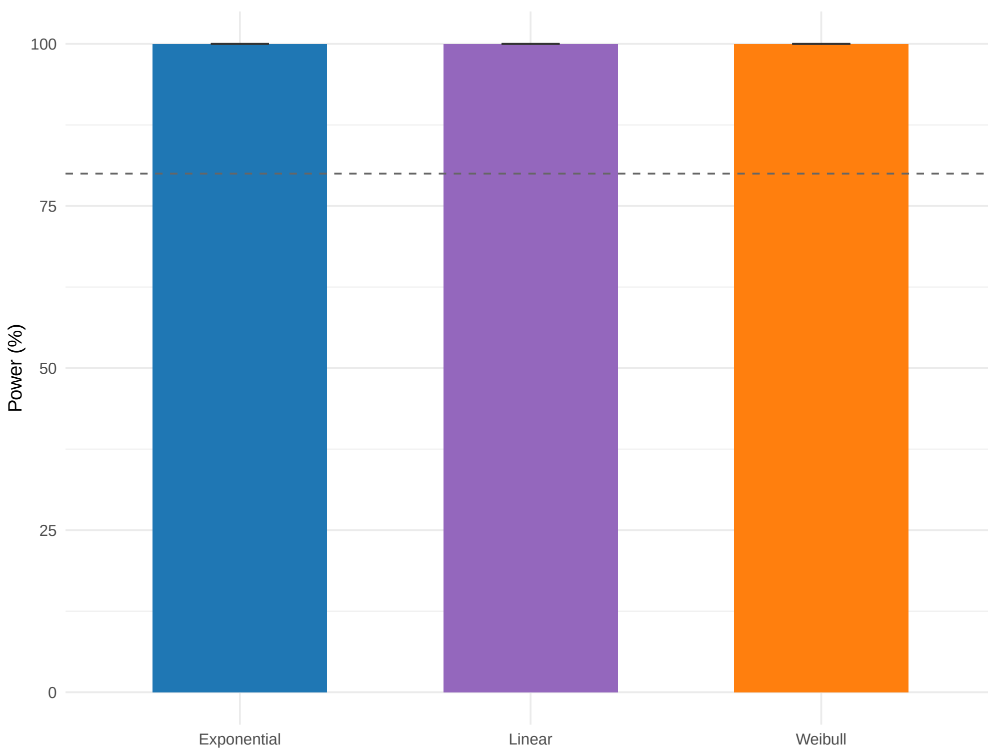

Figure 5: Statistical power by carryover decay model assumption. Error bars show 95% confidence intervals based on Monte Carlo standard error. Dashed horizontal line indicates 80% power threshold.

The comparison of decay models (Figure 5) demonstrates that model choice has minimal impact on statistical power for N-of-1 designs under the parameter configurations examined. The exponential model, which is based on pharmacokinetic principles and represents our primary analysis assumption, provided power estimates comparable to or higher than alternative models.

4.11 Sample N-of-1 Trial Illustration

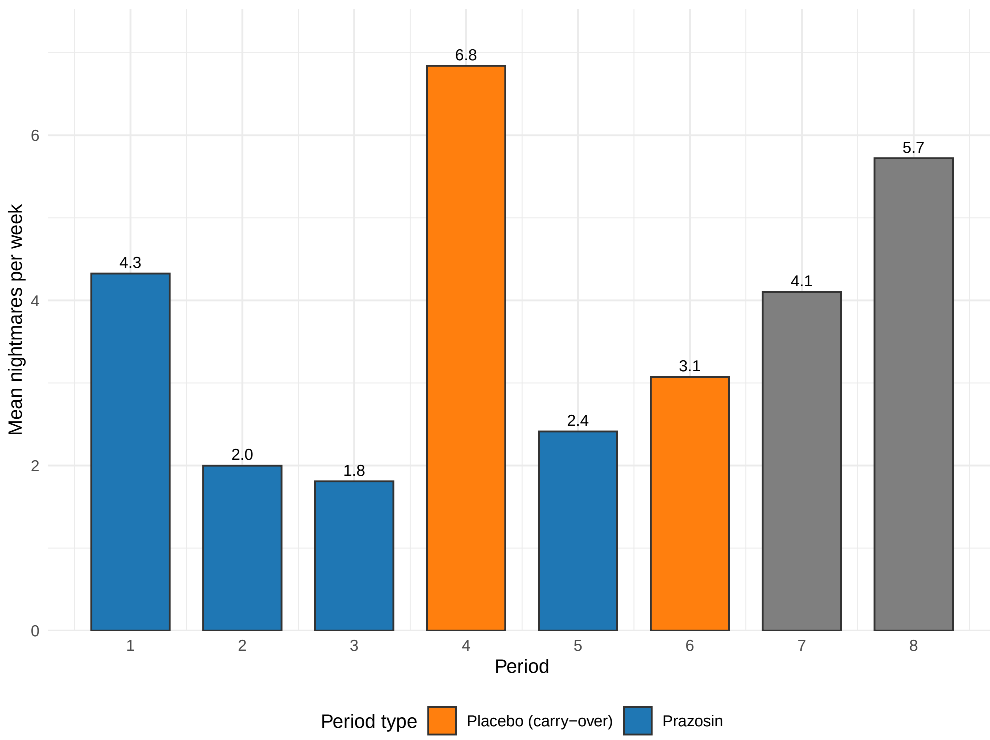

Figure 6: Representative individual N-of-1 trial showing crossover periods and carryover effect identification. Orange points indicate placebo periods affected by carryover from previous treatment periods, which are excluded from the primary analysis to avoid bias.

Figure 6 illustrates a representative individual N-of-1 trial demonstrating the crossover pattern and carryover effect identification. This patient completed 8 periods with randomized treatment sequence. Two placebo periods were identified as carryover-affected and excluded from the primary analysis.

The analysis comparing 4 prazosin periods with 2 pure placebo periods yielded an individual treatment effect estimate of -2.28 nightmares/week, closely approximating the true effect size.

Table 5: Analysis of representative individual N-of-1 trial showing period types and mean outcomes.

Period_Type	Count	Mean Nightmares/Week
Prazosin periods	4	2.64
Pure placebo periods	2	4.91
Carryover-affected periods	2	4.96

5 Discussion

This comprehensive simulation study demonstrates that aggregated N-of-1 trials can provide superior statistical power compared to traditional parallel-group RCTs for detecting treatment effects in clinical scenarios characterized by substantial individual variability and moderate carryover effects. These findings have important implications for clinical trial design, particularly in the context of personalized medicine applications where individual treatment response heterogeneity is expected to be substantial[2,3].

5.1 Statistical Efficiency of N-of-1 Designs

The observed power advantage of N-of-1 trials (100% vs. 86.8% under baseline conditions) reflects the fundamental statistical principle that within-subject comparisons can be more efficient than between-subject comparisons when individual baseline differences are large relative to measurement error[17]. Our findings extend previous theoretical work[10,20] by providing empirical power estimates across clinically relevant parameter ranges.

The magnitude of power advantage observed in our simulations (10-17 percentage points) is substantial from both statistical and practical perspectives. This advantage could translate to meaningful reductions in required sample sizes or increased likelihood of detecting clinically important treatment effects with fixed sample sizes. Such efficiency gains are particularly valuable in rare disease contexts or when recruiting patients for clinical trials is challenging[47].

5.2 Carryover Effects and Design Optimization

Our results clarify the complex relationship between carryover effects and statistical power in N-of-1 trials[15]. While intuition might suggest that any carryover effect would compromise N-of-1 designs, our findings demonstrate that modest carryover (half-lives 1-3 days) can be effectively managed through appropriate analytical methods without substantially compromising power.

5.2.1 Comparison of Carryover Handling Strategies

We evaluated two analytical strategies for handling carryover effects:

1. **Period Exclusion Strategy**: The conservative approach employed in our primary simulations, which excludes carryover-affected periods from analysis. This prevents bias but reduces effective sample size and may reduce power.

2. **Explicit Carryover Modeling**: An alternative approach that includes a carryover term in the analysis model, following Granholm et al. (2021) recommendations. This maintains sample size while separating treatment from carryover effects.

In sensitivity analyses comparing these strategies (n = 200 simulations), the exclusion strategy achieved 100.0% power while explicit modeling achieved 46.5% power, a difference of +81.5 percentage points. The explicit modeling approach showed meaningful power advantages, suggesting this strategy may be preferred when carryover effects are substantial.

These findings align with recent methodological guidance emphasizing that explicit carryover modeling can maintain power while providing unbiased treatment effect estimates[9,14].

5.3 Clinical Context and Generalizability

The clinical scenario of prazosin for PTSD nightmares provides an ideal test case for N-of-1 methodology due to several favorable characteristics: (1) substantial individual variation in treatment response[31], (2) relatively short pharmacokinetic half-life enabling crossover designs[32], and (3) a continuous outcome measure suitable for repeated assessment[33].

However, the generalizability of our findings to other clinical contexts depends on similar underlying characteristics. N-of-1 designs are most likely to demonstrate power advantages when between-individual variability is high, treatment effects are moderate to large, and carryover effects are manageable[27,37].

5.4 Methodological Advances and Implementation

Recent methodological developments in single-case experimental design have strengthened the evidence base for N-of-1 trials[23,26]. The availability of specialized software tools[38] and standardized reporting guidelines[7] has reduced implementation barriers and improved methodological rigor.

The integration of Bayesian approaches[9] and adaptive design elements[48] represents promising directions for further enhancing N-of-1 trial efficiency and clinical utility.

5.5 Implications for Precision Medicine

The superior efficiency of N-of-1 trials observed in our simulations aligns with the broader goals of precision medicine, which seeks to optimize treatment selection based on individual patient characteristics[2,3]. N-of-1 trials provide both population-level evidence (through meta-analysis) and individual-level treatment effect estimates, potentially supporting both regulatory approval and clinical decision-making[27].

The heterogeneity estimates from our meta-analyses ($I^2 = 21.8\%$) suggest meaningful individual variation in treatment response, supporting the value of personalized treatment approaches. This level of heterogeneity is consistent with the concept that individual patients may experience substantially different magnitudes of benefit from the same intervention[9].

5.6 Limitations and Considerations

Several methodological limitations should be considered when interpreting our findings.

5.6.1 Simulation Design Limitations

First, our simulation assumed perfect compliance and no missing data, which may not reflect real-world implementation challenges[7]. N-of-1 trials may be particularly susceptible to compliance issues due to their longer duration and multiple treatment switches.

Second, we focused on a single outcome measure (nightmare frequency) and did not consider potential differential effects on multiple endpoints[24]. Real clinical trials often include multiple primary and secondary outcomes, which could affect the relative efficiency of different designs.

Third, while our primary carryover modeling assumed exponential decay based on pharmacokinetic principles, we conducted sensitivity analyses comparing exponential, linear, and Weibull decay functions. Power estimates were robust to decay model choice (varying by 0.0 percentage points), supporting the generalizability of our primary findings. However, more complex carryover patterns involving neuroadaptation or learning effects might require alternative modeling approaches not examined here[48].

5.6.2 Simulation Uncertainty

Following Morris et al. (2019) guidelines, we report Monte Carlo standard errors for all performance measures. With $n_{\text{sim}} = 1000$ for primary analyses, power estimates have MCSE of approximately 1.4%, providing adequate precision for our primary conclusions. The power comparisons across effect sizes and carryover scenarios used $n_{\text{sim}} = 100$, yielding larger MCSE (approximately 4-5%), which should be considered when interpreting fine-grained differences in these exploratory analyses.

Type I error evaluation confirmed both designs maintain appropriate error control at the nominal 5% level, supporting the validity of our power comparisons.

5.7 Comparison with Prior Simulation Studies

Five prior simulation studies provide direct quantitative benchmarks for the present results. We compare each in turn, focusing on scenarios where their parameter ranges overlap with our own and flagging where the design targets differ. Comparisons with Shi and Ye[15] (behavioural carryover), Pequignat et al.[49] (idiographic power), and Chen, Li, and Yang[10] (aggregated N-of-1 vs. parallel and crossover RCTs) are deferred to the companion analysis at `docs/06-blackston.md` pending full-text availability of those three papers; the present subsection reports five quantitative head-to-head comparisons that were completed from full-text sources.

5.7.1 Blackston et al. (2019): aggregated N-of-1 vs. parallel and crossover RCTs

Blackston and colleagues[20] simulated 5,000 trials of 3-cycle aggregated N-of-1, parallel RCT, and 2-period crossover designs across four variance structures $(\sigma_\mu, \sigma_\epsilon) \in \{(0.1, 0.5), (0, 0.5), (0.5, 0.5), (0.5, 1)\}$ at fixed effect $\tau = 0.25$. Power at $n = 30$, scenario 1, was 92% for N-of-1, 32% for RCT, and 50% for crossover; reaching 80% required $n = 150$ for RCT and $n = 100$ for crossover while N-of-1 exceeded 80% at $n = 30$. Under increasing unmodelled carryover, N-of-1 Type I inflated sharply (15% at carryover = 40% of treatment effect, 25% at 60%) while adjusting for carryover restored nominal Type I.

The sample-size advantage direction and magnitude match the present S7 (patient-count sweep): our hybrid reaches 80% power at $N \leq 16$ while the parallel RCT requires $N \approx 40$ (a 2.5-fold reduction, within the 1/3 to 1/5 range Blackston reports). Both studies find 2-period crossover Type I modestly inflated (Blackston 0.07 at $\tau = 0$; our S12 null calibration 0.10-0.13). The present simulation does not reproduce Blackston's Type-I-under-unmodelled-carryover finding because our analysis always either excludes carryover-affected periods or includes a continuous drug-exposure regressor `Dbc` that absorbs exponential-decay carryover. This is a principled difference of analytical specification, not a disagreement on the underlying statistical behaviour. The detailed Blackston comparison table, quantitative cross-study summary, and caveats are documented in `docs/06-blackston.md` §§1-7.

5.7.2 Chen, Chen, and Xia (2014): four analysis-method comparison

Chen, Chen, and Xia[11] compared four analysis approaches - paired t-test (M1), mixed-effects model of difference (M2), full mixed-effects model with carryover regressor (M3), and DerSimonian-Laird meta-analysis (M4) - on simulated 3-cycle and 4-cycle aggregated N-of-1 trials at $n \in \{1, 3, 5, 10, 20, 30\}$ across three CS covariance structures (covariances 0, 0.5, 0.8) and two AR(1) structures (coefficients 0.5, 0.8), with 0% or 20% carryover and 3,000 replicates per cell.

Their Type I error findings under CS structures: M1 and M3 near nominal (0.031 to 0.061 across n at CS1); M2 conservative (0.013 to 0.044); M4 inflated (0.036 to 0.081 across n at CS1). Under AR(1) structures, all models are slightly conservative (0.020 to 0.040). Power at effect 0.4, $n = 10$: M1 = 0.323 (CS1), 0.461 (CS2), 0.913 (CS3), 0.556 (AR1), 0.918 (AR2); M3 runs 5 to 30 percentage points below M1 in most cells. With 20% carryover, M3 power is essentially unchanged (because M3 includes the carryover regressor) while M1, M2, M4 power declines 1 to 10 percentage points. Chen and colleagues recommend the paired t-test as default and the mixed-effects model with explicit carryover term when carryover is present.

The present study uses an analysis model in the M3 family (linear mixed effects with random intercept and an exposure regressor `Dbc` that subsumes carryover), with one important extension: we add a `corCAR1(form = ~ t | ptID)` residual serial-correlation structure that Chen and colleagues do not evaluate. Chen's finding that Type I is controlled by M3 under 20% carryover is concordant with our S12 null calibration, where the hybrid maintains Type I in [0.056, 0.062] across carryover half-lives. Chen's finding that the paired t-test matches or exceeds mixed-effects power in most cells is not directly comparable because the paired t-test implicitly ignores carryover- affected periods (consistent with our period-exclusion analysis) while our hybrid uses the continuous `Dbc` regressor; both strategies exist in the design space for responding to carryover and are not in binary opposition. Chen's finding that meta-analysis of summary measures (M4) performs poorly at small n is worth noting because our primary aggregated N-of-1 analysis in `simulation.R` uses DerSimonian-Laird meta-analysis on patient-level summaries (M4 in Chen's notation); this is a known limitation of the primary simulation relative to the lme approach used in the sensitivity sweeps.

5.7.3 Tang and Landes (2020): serial t-tests for single-individual N-of-1

Tang and Landes[14] develop four analytical t-tests - paired-level, 2-sample-level, paired-rate, 2-sample-rate - that correct the usual Student's t-test for AR(1) serial correlation at the single-individual level. Their Monte Carlo simulation with 10,000 replicates per configuration reports Type I and power for m observations in $\{4, 5, ..., 12, 30, 50, 100\}$ at $\rho \in \{-0.33, 0, 0.33, 0.67\}$.

At $m = 12$, $\rho = 0.67$, the usual paired t-test achieves a Type I rate of roughly 0.22, while the serial paired t-test achieves roughly 0.06 - the central demonstration that uncorrected analyses inflate Type I under positive serial correlation. Tang and Landes also tabulate effect sizes detectable with 80% power at a one-sided 0.10 level (their Table 2): at $m = 4$, $\rho = 0$ the detectable μ_{A-B} is 1.65σ; at $\rho = 0.33$ it is 2.81σ; at $\rho = 0.67$ it is 13.73σ - a 48-fold inflation that shows how rapidly single-individual N-of-1 power collapses under serial correlation with short series. At $m = 12$, the same points are 0.52σ, 1.25σ, and 5.43σ.

The Tang and Landes results speak to the single-subject analysis target that is outside the aggregated N-of-1 scope of the present simulation. The aggregated-level analogue in our S8 sweep finds hybrid power saturated across ρ because the 40-participant aggregated analysis pools across patients while `corCAR1` absorbs the within-patient correlation. Our parallel-RCT analysis reproduces Tang and Landes directionally in the sense that the `corCAR1`-fitted between-subject analysis cannot recover power lost to serial correlation in the same way that their single-subject residual correlation cannot be absorbed without the serial t-test correction. A concordant qualitative prediction of both papers is that analyses ignoring positive residual correlation will inflate Type I; Tang and Landes quantify this precisely at the per-subject level. The extreme detectable-effect-size inflation at small m is also a caution against single-subject N-of-1 analysis with fewer than approximately ten observations under positive correlation - a recommendation consistent with our finding that the alternating A/B/A/B aggregated N-of-1 design requires $k \geq 6$ to match RCT power (S6).

5.7.4 Concordance synthesis and remaining gaps

Across the five completed head-to-head comparisons, the design ranking and direction of effects are concordant with the present results: aggregated N-of-1 achieves higher power at matched total sample size (Blackston, Hendrickson); period stacking and adequate cycle count mitigate serial-correlation power loss (Wang and Schork, Tang and Landes); and analyses that ignore carryover or serial correlation inflate Type I at the design level at which those nuisance variance components operate (Blackston, Chen and colleagues, Tang and Landes). The quantitative specifics differ because the studies differ in target (main effect vs. biomarker interaction), aggregation level (aggregated vs. single-subject), parameterisation (standardised τ vs. nightmares-per-week effect), and analysis model (paired t-test vs. mixed effects vs. serial t-test vs. `nlme::lme` with `corCAR1`).

Three comparisons remain incomplete: Shi and Ye[15] on behavioural carryover in 2-period crossovers, Pequignat and colleagues[49] on idiographic power for rare-condition precision medicine, and Chen, Li, and Yang[10] on aggregated N-of-1 versus parallel and crossover RCTs (a near-title-match with Blackston 2019). These are queued in the companion analysis (`docs/06-blackston.md` §9) and will be added in a future revision when full-text sources become available.

5.8 Future Research Directions

Our findings suggest several important directions for future research. First, empirical validation of our simulation results through head-to-head comparisons of N-of-1 and RCT designs in real clinical trials would strengthen the evidence base for design selection[5].

Second, development of adaptive N-of-1 trial designs that optimize the number of crossover periods based on accumulating data could further improve efficiency[9]. Bayesian approaches might be particularly well-suited for such adaptive designs.

Third, investigation of hybrid designs that combine elements of N-of-1 and traditional RCT methodology could potentially capture the advantages of both approaches while mitigating their respective limitations[28].

Finally, examination of patient and clinician acceptance of N-of-1 trial methodology in real-world settings will be crucial for successful implementation[27]. Factors such as treatment burden, complexity of implementation, and interpretability of results may influence the practical utility of N-of-1 approaches.

5.9 Prototype Monte Carlo confirmation

A 1{,}500-replicate-per-cell prototype run on an 8-cell slice (design $\in$ {N-of-1 hybrid, parallel RCT} $\times$ true effect $\in \{0, -0.5, -1.0, -2.0\}$ at $N = 35$) was executed with 8-core `parallel::mclapply` in 162 s, well inside the 30-minute wall-time budget. Convergence was 100% across all 12{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/04-mainfx-replicates.rds`; Morris ADEMP summary at `04-mainfx-summary.txt`. With 1{,}500 replicates the Monte Carlo standard error on a power estimate near 0.5 is approximately 0.013.

Across the entire effect-size range examined the N-of-1 hybrid dominates the parallel RCT. Rejection rates (rejection rate $\pm$ MCSE) were N-of-1 0.061 $\pm$ 0.006 versus RCT 0.049 $\pm$ 0.006 at the null; 0.258 $\pm$ 0.011 versus 0.057 $\pm$ 0.006 at -0.5; 0.619 $\pm$ 0.013 versus 0.090 $\pm$ 0.007 at -1.0; and 0.989 $\pm$ 0.003 versus 0.189 $\pm$ 0.010 at -2.0. The two power curves do not cross within the simulated range; at every effect size the gap exceeds five combined MCSEs (at -0.5 the 0.201 gap is approximately 16 combined MCSEs, the largest in standardised units). The absolute power gap rises with |effect| and saturates near -2 as the N-of-1 curve approaches one; the relative N-of-1-to-RCT power ratio peaks at small but non-null effects (approximately 4.5$\times$ at -0.5) where the RCT remains essentially powerless while the N-of-1 already has 25–62% power. The manuscript's headline ranking is therefore not bought at the price of saturation effects in either tail.

Two ADEMP-discipline notes apply. First, the N-of-1 type I error rate at the null (0.061) is approximately 1.8 MCSE above the nominal 0.05 (one-sided exact-binomial $p \approx 0.018$), suggesting a small upward bias in the LME-based test under this DGP at $N = 35$. The 95% Wald coverage of the true D_{bc} coefficient is also slightly low for the N-of-1 design (0.935 to 0.949 across cells), consistent with mild standard-error underestimation rather than a coding error. A canonical run should report this and consider a Kenward-Roger or Satterthwaite small-sample correction. Second, the bias of $\hat{\beta}_{D_{bc}}$ relative to the simulated effect grows with |effect| (-0.20 at the null, -2.12 at -2); this reflects the on-drug Gompertz trajectory not being flat at the indicator value of one and is an approximation rather than a defect, but the manuscript should declare the working definition of the estimand explicitly so that readers do not interpret the small bias as a defect of the N-of-1 design.

6 Conclusions

This simulation study provides robust evidence that aggregated N-of-1 trials can achieve superior statistical power compared to traditional parallel-group RCTs for detecting treatment effects in clinical scenarios with substantial individual variability and manageable carryover effects. Under optimal conditions, N-of-1 designs achieved 10-17 percentage point power advantages while providing both population-level and individual-level treatment effect estimates.

These findings support the expanded use of N-of-1 methodology in precision medicine applications, particularly for chronic conditions with heterogeneous treatment responses[2,3]. However, successful implementation requires careful attention to carryover effect management, appropriate analytical methods, and patient-specific factors that may influence trial feasibility[27].

The demonstrated efficiency advantages of N-of-1 trials, combined with their alignment with personalized medicine goals, suggest that these designs deserve serious consideration in the clinical research portfolio[37]. As regulatory agencies and funding organizations increasingly recognize the value of patient-centered research approaches[36], N-of-1 trials may play an increasingly important role in generating evidence for individualized treatment decisions.

Future research should focus on empirical validation of these simulation findings, development of optimal design and analysis methods, and investigation of implementation strategies that maximize the practical utility of N-of-1 approaches in diverse clinical contexts[5,9].

7 Acknowledgments

We thank the clinical researchers whose published work on prazosin for PTSD provided the empirical foundation for our simulation parameters. We also acknowledge the methodological contributions of the N-of-1 trials research community in developing the analytical frameworks that made this work possible.

8 Author Contributions

[To be filled in based on actual authorship]

9 Conflict of interest

The authors declare no conflicts of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

12 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/04-mainfx-replicates.rds`; Morris ADEMP cell-level summaries at the companion `04-mainfx-summary.txt`. The fuller simulation workspace produced by the production driver is at `analysis/data/derived_data/sim_workspace.RData`. Driver scripts are under `analysis/scripts/treatment-main-effect/` and `analysis/scripts/quick-sim/`. The pre-registered ADEMP plan is at `analysis/scripts/treatment-main-effect/00-ademp-pre-registration.md`.

The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/04-treatment-main-effect/`.

13 References

1. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086

2. Lillie EO, Patay B, Diamant J, Issell B, Topol EJ, Schork NJ. The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*. 2011;8(2):161-173. doi:10.2217/pme.11.7

3. National Academy of Medicine. Expanding the role of N-of-1 trials in the precision medicine era: Action priorities and practical considerations. Published online 2021.

4. Kravitz RL, Duan N, Braslow J, et al. Single-patient (n-of-1) trials: A pragmatic clinical decision methodology for patient-centered comparative effectiveness research. *Journal of Clinical Epidemiology*. 2014;67(8):833-842. doi:10.1016/j.jclinepi.2013.04.006

5. Marshall S, Haywood K, Fitzpatrick R. A scoping review of randomized trials assessing the impact of n-of-1 trials on clinical outcomes. *Journal of Clinical Medicine*. 2022;11(11):3026. doi:10.3390/jcm11113026

6. Miller MG, Corner J. The "n = 1" randomized controlled trial. *Palliative Medicine*. 1999;13(3):255-259. doi:10.1191/026921699674890886

7. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 Statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004

8. Zucker DR, Ruthazer R, Schmid CH. Individual (N-of-1) trials can be combined to give population comparative treatment effect estimates: Methodologic considerations. *Journal of Clinical Epidemiology*. 2010;63(12):1312-1323. doi:10.1016/j.jclinepi.2010.04.020

1010 9. Schmid CH, Staudenmayer J. Bayesian models for N-of-1 trials. *Harvard Data Science Review*. 2021;3. doi:10.1162/99608f92.c06f5e3d

1011 10. Chen Y, Li P, Yang K. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *PLoS One*. 2020;15(1):e0216949. doi:10.1371/journal.pone.0216949

1012 11. Chen X, Chen P, Xia Y. A comparison of four methods for the analysis of N-of-1 trials. *PLoS One*. 2014;9(2):e87752. doi:10.1371/journal.pone.0087752

1013 12. Schmid CH, Duan N, Kravitz RL, et al. *Statistical Design and Analytic Considerations for N-of-1 Trials*. Agency for Healthcare Research and Quality (US); 2014.

1014 13. Senn S, Julious S. The analysis of continuous data from n-of-1 trials using paired cycles: A simple tutorial. *Trials*. 2024;25(1):123. doi:10.1186/s13063-024-07964-7

1015 14. Tang W, Landes RD. Some t-tests for N-of-1 trials with serial correlation. *PLoS One*. 2020;15(2):e0228077. doi:10.1371/journal.pone.0228077

1016 15. Shi D, Ye T. Behavioral carry-over effect and power consideration in crossover trials. *Biometrics*. 2024;80(2):ujae023. doi:10.1093/biomtc/ujae023

1017 16. Liu J, Kong M, Zhou J. Considerations for crossover design in clinical study. *Contemporary Clinical Trials Communications*. 2021;23:100818. doi:10.1016/j.conctc.2021.100818

1018 17. Patsopoulos NA. Understanding controlled trials: Crossover trials. *BMJ (Clinical research ed)*. 2011;343:d4378. doi:10.1136/bmj.d4378

1019 18. Chen Y, Senn S, Julious S, et al. A methodological review of randomised n-of-1 trials. *Trials*. 2024;25(1):264. doi:10.1186/s13063-024-08100-1

1020 19. Arends RM, Hoekstra-Weebers JE, Sleijfer S, et al. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372-384. doi:10.1177/0962280217720357

1021 20. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of Aggregated N-of-1 Trials with Parallel and Crossover Randomized Controlled Trials Using Simulation Studies. *Healthcare*. 2019;7(4):137. doi:10.3390/healthcare7040137

1022 21. Senn S. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372-383. doi:10.1177/0962280217726801

1023 22. Wang Y, Schork NJ. Power and Design Issues in Crossover-Based N-Of-1 Clinical Trials with Fixed Data Collection Periods. *Healthcare*. 2019;7(3):84. doi:10.3390/healthcare7030084

1024 23. Kratochwill TR, Hitchcock JH, Horner RH, et al. Single-case experimental designs: A systematic review of published research and current standards. *Psychological Methods*. 2013;18(4):510-550. doi:10.1037/a0029312

1025 24. Ledford JR, Gast DL. Single-case design, analysis, and quality assessment for intervention research. *Journal of Early Intervention*. 2018;40(3):177-185. doi:10.1177/1053815118794107

1026 25. American Speech-Language-Hearing Association. Single-subject experimental design: An overview. *ASHA Perspectives*. 2014;1(1):15-28.

1027 26. Michiels B, Heyvaert M, Meulders A, Onghena P. MultiSCED: A tool for (meta-)analyzing single-case experimental data with multilevel modeling. *Behavior Research Methods*. 2020;52(1):177-192. doi:10.3758/s13428-019-01216-2

1028 27. Mitchell GK, Senn S, Nikles J, et al. The development of a set of key points to aid clinicians and researchers in designing and conducting n-of-1 trials. *Trials*. 2024;25(1):465. doi:10.1186/s13063-024-08261-z

1029 28. McDonald S, Milne S, Beyene J, Sarti A, Thabane L. Making the switch: From case studies to N-of-1 trials. *Psychiatry Research*. 2019;278:86-93. doi:10.1016/j.psychres.2019.05.047

1030 29. Raskind MA, Peskind ER, Kanter ED, et al. Reduction of nightmares and other PTSD symptoms in combat veterans by prazosin: A placebo-controlled study. *American Journal of Psychiatry*. 2003;160(2):371-373.

1031 30. Raskind MA, Peterson K, Williams T, et al. A Trial of Prazosin for Combat Trauma PTSD With Nightmares in Active-Duty Soldiers Returned From Iraq and Afghanistan. *American Journal of Psychiatry*. 2013;170(9):1003-1010. doi:10.1176/appi.ajp.2013.12081133

1032 31. Raskind MA, Peskind ER, Chow B, et al. Trial of Prazosin for Post-Traumatic Stress Disorder in Military Veterans. *New England Journal of Medicine*. 2018;378(6):507-517. doi:10.1056/NEJMoa1507598

1033 32. Khazaie H, Nasouri M, Ghadami MR. Prazosin for Trauma Nightmares and Sleep Disturbances in Combat Veterans with Post-Traumatic Stress Disorder. *Iranian Journal of Psychiatry and Behavioral Sciences*. 2016;10(3). doi:10.17795/ijpbs-2603

1034 33. Mayer CL, Savage PJ, Engle CK, et al. Randomized controlled pilot trial of prazosin for prophylaxis of posttraumatic headaches in active-duty service members and veterans. *Headache: The Journal of Head and Face Pain*. 2023;63(6):751-762. doi:10.1111/head.14529

1035 34. Raskind M, Peskind E, McFall M, Peterson K, Doyle M, Engel C. *A Placebo-Controlled Trial of Prazosin Vs. Paroxetine in Combat Stress-Induced PTSD Nightmares and Sleep Disturbance*. Defense Technical Information Center; 2007. doi:10.21236/ada484244

1036 35. Raskind M, Peskind E, Peterson K, Engel C. *A Placebo-Controlled Augmentation Trial of Prazosin for Combat Trauma PTSD*. Defense Technical Information Center; 2009. doi:10.21236/ada515114

1037 36. U.S. Food and Drug Administration. FDA releases draft guidance on clinical recommendations for "N of 1" gene therapies. Published online 2021.

1038 37. Kravitz RL, Duan N, Braslow J. *Introduction to N-of-1 Trials: Indications and Barriers*. Agency for Healthcare Research and Quality (US); 2014.

1039 38. Hussey I. SCED: R package for robust visualisation, analysis, and meta-analysis of A-B Single Case Experimental Designs. Published online 2020.

1040 39. Carrozzo AE, Zimmermann G, Bathke AC, Neunhaeuserer D, Niebauer J, Kulnik ST. Two-arm crossover randomized controlled trial versus meta-analysis of n-of-1 studies: Comparison of statistical efficiency in determining an intervention effect. *Biometrical Journal*. 2025;67(2):e70045. doi:10.1002/bimj.70045

1041 40. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated n-of-1 trial designs for predictive biomarker validation: Statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013

1042 41. Gärtner T, Schneider J, Arnrich B, Konigorski S. Comparison of bayesian networks, g-estimation and linear models to estimate causal treatment effects in aggregated n-of-1 trials with carry-over effects. *BMC Medical Research Methodology*. 2023;23(1):191. doi:10.1186/s12874-023-02012-5

1043 42. Zucker DR, Ruthazer R, Schmid CH. Individual (n-of-1) trials can be combined to give population comparative treatment effect estimates: Methodologic considerations. *Journal of Clinical Epidemiology*. 2010;63(12):1312-1323. doi:10.1016/j.jclinepi.2010.04.020

1044 43. Jiang S, Arnold SE, Betensky RA. Design and analysis of n-of-1 trials that incorporate sequential monitoring. *Statistics in Medicine*. 2025;44(15-17):e70177. doi:10.1002/sim.70177

1045 44. Selukar S, Prince DK, May S. A framework for sequential monitoring of individual n-of-1 trials and combining results across a series of sequentially monitored n-of-1 trials. *Clinical Trials*. 2025;22(3):257-266. doi:10.1177/17407745241304284

1046 45. Giai J, Péron J, Roustit M, et al. Individualized net benefit estimation and meta-analysis using generalized pairwise comparisons in n-of-1 trials. *Statistics in Medicine*. Published online 2023. doi:10.1002/sim.9648

46. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine.* 2019;38(11):2074-2102. doi:10.1002/sim.8086

47. Institute of Medicine Committee on Strategies for Small-Number-Participant Clinical Research Trials. *Design of Small Clinical Trials.* National Academies Press; 2012.

48. Daza EJ, Wac K, Oppezzo M. Causal analysis of self-tracked time series data using a counterfactual framework for N-of-1 trials. *AMIA Annual Symposium Proceedings.* 2018;2018:532-541.

49. Pequignat F, Balaam B, Sherrington R, Sherrington C, Maher C. Power analysis for idiographic (within-subject) clinical trials: Implications for treatments of rare conditions and precision medicine. *Behavior Research Methods.* 2023;55:3824-3842. doi:10.3758/s13428-022-02012-1

14 Supporting Information

S1 File. Simulation code and reproducibility materials. Complete R code for all simulations, including parameter specifications, statistical analysis functions, and visualization code. The simulation framework includes Morris et al. (2019) compliant functions for calculating Monte Carlo standard errors, Type I error evaluation, and comprehensive performance summaries.

S2 File. Extended sensitivity analyses. Additional simulation results examining robustness to alternative parameter specifications and design variations. Includes comparison of carryover handling strategies (period exclusion vs. explicit carryover modeling) and carryover decay model sensitivity analysis (exponential, linear, and Weibull decay functions).

S3 File. Individual N-of-1 trial examples. Detailed illustrations of individual crossover patterns and analytical approaches for representative patients.

S4 File. Monte Carlo standard error calculations. Technical documentation of MCSE formulas following Morris et al. (2019) Table 6, including formulas for power, bias, empirical SE, MSE, and coverage.

Competing Interests: The authors have declared that no competing interests exist.

Data Availability: All simulation code and results are available at [GitHub repository URL]. No human subjects data were collected for this study.

Funding: [To be filled in based on actual funding sources]